

Ramshankar Bhuvaneshwaran

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Education

Northeastern University

M.S. Information Systems

Anna University

B.E. Computer Science and Engineering

Boston, MA

Expected Dec 2025

Chennai, India

Jun 2019 – May 2023

Skills

Languages: Python, SQL, PySpark, R, Shell Scripting, FastAPI, Flask, CUDA, Triton, HTML, CSS, JS

ML/DL: TensorFlow, Keras, PyTorch, Scikit-learn, NLTK, XGBoost, Pandas, NumPy, ggplot

Data Engineering & GenAI : Spark, Kafka, Airflow, AWS Glue, Snowflake, RAG, Graph RAG, Fine-tuning, AutoGen, Langchain

Cloud & ML Ops: AWS (SageMaker, EMR, S3), Docker, A/B Testing, Kubernetes, AWS pipelines, Bedrock

Databases & Vector DB: Postgres, MongoDB, Redis, MySQL, Neo4j, ChromaDb

Experience

AI Engineer - Consultant

May 2024 – Aug 2024

BulkBeings

Chennai, India

- Full Fine-tuning foundational LLMs (Mixtral, Qwen, Llama) using FSDP parallelization on GCP A100s, implementing hybrid SFT-DPO training for context-aware detection models with **65K+ synthetic samples** generated via AWS Bedrock.
- Implemented intensive filtering on public datasets for sparse feature selection, removing **40%** unrelatable conversations; built ETL with AWS Glue/PySpark to process and transform **public conversational text dataset (200GB+)**, reducing processing time by **35%**.
- Conducted A/B testing framework for newer LLM variants, for gathering metrics on prediction accuracy and to compare our model.

Research and Development Engineer

May 2023 – Dec 2023

BulkBeings

Chennai, India

- Prototyped and productionalized OCR pipeline (ViT+CRNN) achieving **98% accuracy**, while making the model **42% faster** via ONNX/CUDA optimization; deployed on AWS L4 EC2 instance, API gateway with **less than 100ms p99 latency**.
- Developed two-stage Conv1D-Transformer model for ECG classification (**93% F1-score**, 10 cardiac conditions), implementing feature engineering for temporal/long-range patterns with dynamic windowing algorithms for patient-specific heart rates
- Built retraining pipelines on AWS SageMaker with real-time monitoring dashboards tracking accuracy and data drift metrics.

ML Engineer Intern

Jan 2022 – Aug 2022

Velocity Global Solutions Pvt

Chennai, India

- Architected end-to-end ML pipelines in AWS, implementing predictive segments (high-value, at-risk) using XGBoost on **behavioral patterns** and validating clusters against campaign response data, resulting in a **45% increase** in campaign conversion rates.
- Led cross-functional team in developing retail mix optimization system for a supermarket chain in India using Bayesian hierarchical models (PyMC3), increasing recurring customer LTV prediction accuracy by **34%**.

Key Achievements - Critical Problem Resolution

Data Quality Crisis: Discovered **20% corrupted samples** in public conversational text datasets; designed intensive filtering pipeline for sparse feature extraction using encoder models and built synthetic data generation with dialect/slang variations.

Medical Annotation Error Detection: Identified critical misannotations in millisecond level affecting **15% of proprietary ECG dataset** through custom visualization tool; led initiative with radiologists for systematic re-annotation, preventing potential medical misdiagnosis in dataset.

Compute Cost Optimization: Noticed **70% redundant model calls** for recurring documents in OCR pipeline; architected caching system for frequent bounding boxes, reducing monthly GPU costs by **\$12K** and inference latency by **42%**.

Projects

Distributed Text-to-Image Generation with Cost Optimization | *PyTorch, Dask, DDPM, DDP* 🔄

- Implemented DDPM with U-Net architecture; evaluated CPU parallelization (Multiprocessing, Joblib, Dask) vs GPU strategies to reduce preprocessing costs by **60%**.
- Accelerated training via Automatic Mixed Precision (AMP) with Exponential Moving Average (EMA) weight averaging ($\beta = 0.999$), reducing training time by **11%** ($23.5 \rightarrow 20.9$ hrs/epoch) and loss variance by **29%** through gradient histogram stabilization.

SEC Financial Data ETL Platform | *Snowflake, Airflow, S3, DVT* 🔄

- Built scalable ETL for SEC financial statements; implemented 3-tier storage (Raw/JSON/Denormalized) with Data Validation Tool validation ensuring **99.9% schema integrity**.
- Engineered automated ingestion pipelines via Airflow/S3 for Balance Sheet, Income Statement, Cash Flow data; deployed interactive query interface reducing analyst query time by **70%** combining **Streamlit** for visualization and **FastAPI** backend services.

Llama3 Pure C/CUDA Implementation | *C, CUDA* 🔄

- Optimized SwiGLU kernel with bfloat16/Packed128 achieving **15% forward/5% backward speedup**; implemented RMSNorm via cooperative groups with warp-level primitives, reducing latency **57%/88%** (fwd/bwd) across all transformer layers.
- Enhanced memory coalescing/shared memory across attention, RoPE, SwiGLU kernels using texture memory and L2 cache optimization; achieved **76% latency reduction** and **85%** GPU utilization for production-scale deployment.